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What Are the Qualities of “Good” Malware Datasets?

Investigators:

The University of Texas at Dallas

Team 2

Sai Tharun Reddy Mulka - Student researcher - sxm220351@utdallas.edu

Mohammad Rafeian - Student Researcher - mxr230010@utdallas.edu

Pranathi Munnangi - Student - pxm220084@utdallas.edu

Sahithi Reddy Mopuru - Student - sxm220353@utdallas.edu

Sri Kruthi Madugula - Student - sxm220429@utdallas.edu

Brian Ricks - Faculty Support - absolutefunk@utdallas.edu

Problem Mentor:

Frank Posluszny - MITRE - malwarefrank@mitre.org

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1. Executive Summary

The primary goal of this research project is to critically assess existing methodologies for constructing malware datasets and identify the key features that make a dataset both effective and realistic for both research and practical cybersecurity applications. Current malware datasets are often biased, primarily due to the convenience of collecting certain types of data, leading to an inaccurate representation of real-world malware behavior. This bias significantly impairs the validity of research findings and limits the applicability of results beyond controlled environments. By understanding these biases and limitations, the project aims to establish best practices for creating more comprehensive and representative datasets.

To achieve this, the project adopts a multifaceted approach that includes a systematic literature review and cross-disciplinary insights from other fields that deal with complex data, such as weather prediction and IoT. By examining these fields, which share characteristics with malware research—such as the need for datasets that are regularly updated and the unpredictability of features—the project aims to develop a robust framework for constructing malware datasets that better capture the diverse behaviors of real-world threats. These cross-industry comparisons will provide valuable insights into how other fields tackle similar challenges, ultimately guiding the development of more versatile malware datasets.

The project begins by reviewing the limitations identified in existing malware dataset papers to pinpoint the factors that hinder the generalization of these datasets across different research projects. This initial step helps to uncover why many datasets are not broadly applicable and highlights areas for improvement. Following this, the research synthesizes recommendations from the literature to define the qualities that make a "good" malware dataset. These insights will inform the creation of guidelines and best practices for future datasets, ensuring they are comprehensive, diverse, and representative of a wider range of malware behaviors.

Through a combination of literature review, feature extraction, and cross-industry comparison, the project seeks to provide the cybersecurity community with a comprehensive set of guidelines for building more effective malware datasets. These guidelines will be grounded in an understanding of the unique challenges that malware researchers face and will incorporate lessons learned from other domains. The ultimate goal is to enhance the quality, reliability, and applicability of malware datasets, paving the way for more accurate research and stronger defenses against evolving cyber threats.

2. Introduction

2.1. Top-Level Research Project Description/ Problem Statement

This project aims to establish the key characteristics of a "good" malware dataset by thoroughly examining the limitations and biases present in existing datasets. Many current datasets are biased due to the ease of collecting certain types of data, which leads to an inaccurate representation of real-world malware behaviors and characteristics. This skewed data can significantly affect the validity of research findings, making it difficult to generalize results beyond controlled environments.

To address these challenges, the project will conduct an in-depth evaluation of current practices in malware dataset construction, drawing from research in both cybersecurity and other fields where data realism and diversity are critical. By leveraging cross-disciplinary insights, the project seeks to develop a set of best practices and guidelines for building malware datasets that accurately capture the range of features and behaviors exhibited by real-world threats. The goal is to create datasets that are comprehensive, diverse, and applicable across various research and practical scenarios, thereby enhancing the quality and reliability of future malware studies.

2.2. Purpose Statement

The primary goal of this project is to critically evaluate the current methodologies used to construct malware datasets and to identify the characteristics that make a dataset both realistic and effective for research and practical applications. This involves a deep dive into existing literature and practices in malware research as well as other disciplines that deal with complex, real-world data. For more advanced student groups, the project may also include conducting a survey among malware analysis researchers and practitioners to gather first-hand insights on the qualities they believe are essential in a "good" malware dataset. The ultimate aim is to produce a comprehensive set of guidelines and recommendations that can be used by the research community to develop more accurate, unbiased, and representative malware datasets.

Identifying Limitations of Malware Datasets

Our first task was to review the limitations highlighted in malware dataset papers. This step was crucial for identifying the characteristics that make these datasets less applicable across different projects, helping us understand what prevents them from being generalized.

Extracting Recommendations for Malware Datasets

Next, through a careful literature review, we gathered recommendations on the qualities that define good malware datasets. These recommendations provide insights into how future datasets can be improved for broader use in cybersecurity.

Expanding Insights from Other Fields

In addition to cybersecurity, we examined other fields like weather prediction to extract recommendations for dataset qualities. These fields share characteristics with malware datasets, such as the need for regular updates and unpredictability of future features. We argue that by incorporating recommended qualities from these fields, we can design more robust and versatile malware datasets.

This approach of cross-industry evaluation provides a more holistic view of how to overcome existing limitations and improve dataset generalization in malware research.

2.3. Motivation

The motivation behind this project stems from the growing challenges in malware research, where existing datasets often fail to accurately represent real-world threats due to biases and limitations in their construction. As malware continues to evolve in complexity and sophistication, having reliable and diverse datasets is crucial for developing effective detection and defense mechanisms. By addressing the gaps in current dataset practices, this project aims to contribute to the creation of more realistic and generalizable datasets, thereby improving the quality of malware research and enhancing cybersecurity outcomes.

Furthermore, by drawing insights from other fields that deal with complex, unpredictable data, this project seeks to introduce innovative strategies for dataset construction. This cross-disciplinary approach will help ensure that future malware datasets are not only comprehensive but also adaptable to the dynamic nature of cyber threats, making them more useful for researchers and practitioners in the fight against malware.

3. Literature Review Summary

In conducting the literature review for this project, a total of 40 papers were investigated across various industries to identify key insights into dataset construction and characteristics that can be applied to the creation of more effective malware datasets. The majority of the papers—16 in total—focused on cybersecurity, providing a foundational understanding of the current state of malware datasets, their limitations, and the best practices for their development. These studies highlighted various biases, such as those arising from convenience sampling, and offered recommendations on how to ensure datasets are more representative of real-world threats.

In addition to cybersecurity, the review extended to other fields where dataset construction plays a critical role in research and practical applications. Four papers from the machine learning, image processing, and graph machine learning domains provided valuable insights into the use of diverse, dynamic, and regularly updated datasets, which are essential for improving the robustness of models in these areas. Three papers from the building and compiler industries offered perspectives on how datasets are structured for applications involving complex systems,

which share challenges with malware dataset construction in terms of unpredictability and the need for real-time data updates.

The literature also included studies from the Internet of Things (IoT) and network domains (6 papers), which provided useful parallels to malware datasets in terms of the need to capture diverse behaviors and the rapid evolution of data characteristics. A single paper from weather prediction underscored the importance of datasets that reflect long-term variability, while five papers from power and energy prediction offered lessons on handling continuous and real-time data that is subject to change. Finally, two papers from the hardware industry were reviewed, which discussed the design of specialized datasets for systems with unique constraints, offering insights into how tailored datasets can improve accuracy and utility in complex domains like cybersecurity. Overall, the literature review highlighted the value of cross-industry insights in overcoming common challenges in dataset construction and improving the representativeness of future malware datasets.

4. Method and Procedures

4.1. Characterization of Methods Used

Our research follows a qualitative methodology that includes:

Systematic Literature Review: We examined a wide range of research papers on malware datasets, compiling data on limitations and recommendations.

Comparative Analysis: We evaluated how other fields, which deal with similar challenges (e.g., unpredictability, constant updating), structure and generate their datasets.

Synthesis of Findings: The final step involves synthesizing insights from both cybersecurity and other fields to propose a set of best practices and recommendations for future malware datasets.

4.2. Procedures / Plans

Our research is structured into three key tasks:

Literature Review: We conducted an extensive review of over 40 papers, focusing on the characteristics of malware datasets and the qualities that define good datasets across various fields.

Feature Extraction: From the reviewed papers, we extracted the strengths, weaknesses, limitations, and recommended features of both malware datasets and datasets from related fields.

Cross-Industry Comparison: We analyzed the dataset recommendations from fields like weather prediction, image processing, and IoT, drawing parallels to the cybersecurity domain, to identify transferable qualities that could enhance malware datasets.

4.3. Schedule

Phase	Date	Programming Group
	August 27	Project Bids
	August 30	Project Assignments
	September 8	Project Proposal
Phase 1	September 9 - September 15	Initial research on existing malware datasets and their limitations
Phase 2	September 16 - September 31	Systematically review research papers from cybersecurity and related fields. Identify and categorize the limitations, strengths, and recommendations for malware datasets.
Phase 3	October 1 - October 14	Extract key characteristics that define good malware datasets from the reviewed papers. Focus on features like temporal data handling, dataset diversity, and representativeness.
Phase 4	October 8 - October 14	Analyze cross-disciplinary datasets, identifying applicable techniques and qualities from fields like atmospheric sciences and image processing.
Phase 5	October 15 - October 21	finding good practices for creating datasets/ malware datasets
	October 25 - October 27	Midterm presentations / Midterm deliverables

Phase 6	October 28 - November 10	Synthesize findings from the literature review and comparative analysis to develop best practices for malware dataset creation.
Phase 7	November 11 - November 27	Review and refine the best practices based on feedback. Address any gaps or additional considerations before finalizing deliverables.
	November 28	Final report writing and presentation practice run

4.4. Deliverables

We have three deliverables:

Spreadsheet: A Google spreadsheet compiling the literature review, feature extraction, and cross-industry analysis. It includes findings from 40 research papers, key characteristics for effective datasets, and techniques from other fields.

Report: A detailed document summarizing the findings, methodology, insights, and recommendations for improving malware datasets, based on the literature review and comparative analysis.

Presentation: Slides highlighting the project's key results, proposed best practices, and recommendations for future malware dataset development.

4.5. Limitations and Delimitations

Limitations:

This project is constrained by the availability and diversity of existing literature and publicly accessible malware datasets. Many datasets are proprietary, limiting a comprehensive review of all potential biases and limitations. Additionally, the project does not directly create or test new malware datasets but rather focuses on analyzing existing methodologies and frameworks. This theoretical approach may limit its immediate applicability until further empirical testing is conducted.

Delimitations:

The study intentionally focuses on cross-disciplinary comparisons with fields like weather prediction and IoT to extract broader dataset recommendations. However, it does not delve into the technical specifics of malware detection techniques or algorithm development. The scope is also confined to academic and practical applications in malware dataset design, excluding other cybersecurity aspects such as network security or incident response.

5. Findings Overview

5.1. Findings Overview

The study identifies critical characteristics for building high-quality malware datasets by analyzing cybersecurity and other industries such as IoT, weather prediction, image processing, and compilers. Current malware datasets are often limited by narrow scope, lack of diversity, outdated samples, and biases, hindering their utility in real-world scenarios. Through a literature review of 30+ papers and cross-disciplinary comparisons, this research highlights essential qualities like temporal coherence, realistic construction, and frequent updates. Recommendations include ensuring datasets capture a wide variety of malware types, behaviors, and attack scenarios while maintaining data integrity and addressing biases. Insights from fields like weather prediction emphasize the need to handle rare but impactful events, while IoT research underscores the importance of traffic balance and protocol-specific features. By integrating these findings, the proposed framework enables the creation of robust, scalable, and generalizable datasets suitable for academic and operational malware research.

5.2. Detailed Findings

In this section, we present the detailed findings of limitations and recommendations across various disciplines, including Cybersecurity, Image Processing and GraphML, Building and Compilers, IoT and Networks, Atmospheric Sciences and Weather Prediction, Energy Consumption, and Hardware.

Cybersecurity

This section examines challenges in cybersecurity, focusing on historical coherence, dataset bias, class imbalance, model degradation over time, and the trade-offs between static and dynamic analysis methods.

Characteristic	Description	Limitations/Considerations
Historical Coherence	Training data should precede test data chronologically to avoid bias.	Using future data for training can lead to inflated performance results.
Random Selection Bias	Avoid randomly selecting training data without considering the timeline.	May incorporate "future" malware in training, skewing results.

Model Degradation Over Time	Detection model performance decreases as the gap between training and test datasets increases.	Older training data may not represent new malware threats.
Diverse Datasets	Training on a broad set of malware families improves generalization.	Limited diversity in training data can result in poor performance on unseen or rare threats.
Realistic Construction	Datasets should mimic real-world timelines and conditions of malware appearance.	Unrealistic datasets can lead to poor real-world performance.
Measure Multiple Performance Metrics	Evaluate metrics beyond runtime, such as detection accuracy, false positives/negatives, and memory usage.	Focusing solely on runtime can neglect important aspects of model effectiveness.
Raw and Processed Features	Raw features allow custom feature engineering, while processed features help focus on algorithm optimization.	Using only one type may limit experimentation or model performance.
Realistic Validation('In the Wild')	Real-world testing ensures the model's effectiveness outside the lab environment.	Lab settings may not cover all scenarios encountered in the wild.
Select Informative Features	Focus on features that add value to detection while avoiding overfitting and high computational costs.	Including too many features can lead to overfitting or slow performance.
Handling Adware	Adware detection is challenging due to its ambiguous nature, lying between benign and malicious categories.	Can affect model accuracy; careful labeling and feature selection are needed.
Static Analysis	Extracts features from Portable Executable (PE) files without running the code (e.g., section size, entropy).	Struggles with packed/obfuscated malware; techniques like encryption can hinder feature extraction.

Dynamic Analysis	Observes malware behavior in execution (e.g., system calls, network activity).	Limited execution paths may not reveal all behaviors; anti-analysis techniques may evade detection.
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Image Processing and GraphML

The **GraphML** the key findings are :

1. **Scalability Challenges:** Commonly used graph-based datasets tend to be much smaller than the graphs encountered in real-world applications, limiting the evaluation of models for large-scale scenarios. There is a need for datasets that are sufficiently large to enable training on complex models and ensure their ability to generalize effectively. The OGB addresses this by offering datasets of various sizes, ranging from smaller graphs to those with up to 100 million nodes, thereby supporting scalability testing across different scales.
2. **Lack of Diversity:** Many datasets lack variety in structural attributes such as node degrees, clustering coefficients, and other graph characteristics, which restricts a model's ability to generalize across different domains. The OGB datasets strive to encompass graphs from a broad spectrum of domains, including social and information networks, biological networks, molecular structures, source code abstract syntax trees (ASTs), and knowledge graphs. This extensive range of domains supports the creation and evaluation of versatile models, allowing for a clear distinction between general-purpose approaches and domain-specific methods.
3. **Inconsistent Evaluation Protocols:** The absence of standardized evaluation protocols across studies has led to inconsistent results, making comparisons challenging. Using fixed data splits and unified evaluation metrics can standardize the evaluation process, improve reproducibility, and allow for more meaningful comparisons of different approaches.
4. **Data Splitting Methods:** Random dataset splitting is commonly used but does not accurately reflect real-world scenarios, where time-based or structurally informed splits are more applicable. OGB employs more realistic splitting methods (e.g., time-based or molecular structure-based splits) to better simulate real-world conditions, providing more practical and meaningful evaluation scenarios.
5. **Balanced Representation:** Balanced datasets are crucial for preventing bias in model training and achieving fair evaluations. Some OGB datasets, particularly those used in molecular studies, are curated to address class imbalance challenges, and evaluation metrics like Average Precision (AP) are employed to account for skewed data distributions.

These findings underscore the necessity for enhanced benchmarks that reflect real-world applications, facilitate reproducible evaluations, and provide consistent frameworks for machine learning research in the context of graph data.

In **Image processing** the key findings are :

The key findings from the paper on good characteristics for malware datasets include:

1. **Realism and Practical Relevance:** Datasets should represent realistic scenarios to ensure practical applicability in real-world situations. Simulated environments or overly controlled datasets may not adequately prepare models for real-life conditions.
2. **Bias Mitigation:** It's essential to identify and minimize biases in the dataset, such as selection, capture, and negative set biases. Ensuring diverse sampling from various sources can help reduce these biases.
3. **Cross-Dataset Generalization:** A high-quality dataset should enable generalization across different datasets, showing minimal performance drop when tested in different environments. This characteristic helps validate the dataset's representativeness of the broader visual world.
4. **Negative Set Inclusion:** Incorporating diverse and realistic negative samples (instances where malware is not present) is crucial for classifier training, ensuring models do not become overconfident and can better distinguish between positive and negative cases.
5. **Training Data Sufficiency:** The dataset should include enough training examples to allow models to learn effectively. Ideally, the dataset size should be large enough to avoid overfitting while capturing the diversity in malware characteristics.
6. **Annotation Quality:** The quality and consistency of labels (annotations) in the dataset significantly affect model performance. Consistent, high-quality annotations ensure accurate training and evaluation.
7. **Data Augmentation and Transformation:** To address capture biases, applying data transformations such as flipping, jittering, or cropping can help create a more robust training dataset.

Building and Compilers

In the field of malware analysis, a notable limitation is that most existing datasets are tailored for specific environments and operating systems, primarily focusing on Windows. This specificity hampers reproducibility across diverse platforms, highlighting the need for comprehensive datasets that encompass various environments, including Windows, Linux, and macOS. Additionally, hardware specifications play a crucial role in the build process and data generation, particularly when utilizing tools like Intel Pin, which analyzes binaries reliably only on Intel CPU-based systems. While Intel Pin is effective for analyzing Windows malware, its reliance on Intel CPUs necessitates re-implementing analysis tools like EvDetector to accommodate other CPU architectures, ensuring broader applicability and reproducibility of findings.

Moreover, the generation of binary files is influenced by different compilers and optimization flags; however, these critical details are not embedded within the binaries themselves. Understanding compiler and flag configurations is essential for identifying vulnerabilities, rewriting binary code, and optimizing energy consumption. Recent studies have proposed datasets that include binary files alongside their associated compiler and optimization flags.

These datasets are employed to train neural networks for predicting optimization levels across various architectures. The authors emphasize the importance of including features such as architecture, compiler, and compiler flags in dataset design, enabling more effective training and improved model accuracy.

Feature	Importance	Application
Environment	Datasets often focus on specific OS (e.g., Windows), limiting reproducibility.	Need for datasets covering various OS (Windows, Linux, macOS) for broader applicability.
Hardware Specs	Intel Pin analyzes binaries reliably on Intel CPUs, limiting analysis to Intel architecture.	Re-implementation of tools (e.g., EvDetector) is required for compatibility with other CPUs.
Compiler	Different compilers produce varied binary outputs.	Essential for vulnerability discovery and performance analysis.
Compiler Flags	Flags can determine binary performance and energy consumption.	Important for optimizing binary rewriting and improving runtime efficiency.
Architecture	Affects how binaries are executed and analyzed.	Critical for training models that predict optimization levels across different architectures.

IOT and Networks

In the context of IoT and network security, these key findings highlight how the dataset contributes to improving accuracy in machine learning models for intrusion detection and what kind of features are critical, especially for attack detection.

1. Dataset Design and Balance for Accuracy

- Issue with Traffic Dominance: A critical finding is that the dataset includes an overwhelmingly large volume of legitimate traffic compared to malicious traffic. This imbalance can skew accuracy metrics, as models may primarily learn from the abundant legitimate traffic, leading to potential misclassification of rare attack types.
- Balancing legitimate and malicious traffic is crucial for ensuring that the models do not overfit to benign patterns and can generalize to detect rare attack instances effectively.

Datasets like BoT-IoT and TON_IoT show that scalable datasets with long-term traffic monitoring improve the detection of low-traffic, high-persistence attacks (e.g., data exfiltration, botnet communications).

2. Feature Selection and Importance

- In IoT datasets like MQTTset, N-BaloT and IoT-23 highlight flow-level features and DNS analysis for botnet detection, MQTTset focuses on timing and message types for detecting DoS and SlowITe attacks. the features extracted focus heavily on the characteristics of the MQTT protocol, including both network-layer features (TCP/IP) and application-layer MQTT features. These include flags, session information, message types, QoS levels, Timing-Based, Traffic Length, Size and other protocol-specific data.

3. Attack-Specific Features and Detection

- Each type of attack has specific protocol-related features that are critical for detection. Flooding attacks rely on traffic volume and message repetition features, while more subtle attacks like SlowITe depend on timing and session management features. While designing a dataset targeting attack specific features will help the model to detect well.

4. Machine Learning Model Performance

- Ensembled based approach gave good results, It worked well in detecting anomalies by analyzing multi-dimensional features like message types, QoS levels, and timing differences. BoT-IoT employed traditional machine learning models like random forests and gradient boosting supports the usage of ensemble models. For time-series anomaly detection TON_IoT showed RNN, Autoencoders worked very well.

Atmospheric Sciences and Weather prediction

This section highlights the importance of diverse datasets, rare event inclusion, and key feature selection to improve model accuracy and adaptability in weather prediction.

1. Data Diversity and Representativeness

In both atmospheric sciences and weather prediction , datasets need to capture a wide range of environmental conditions, geographic regions, and time periods to ensure models can generalize across different weather patterns, including extreme events. This diversity allows for more accurate and robust predictions. Similarly, a good malware dataset** should encompass a variety of malware types, attack strategies, operating systems, and real-world scenarios. This ensures the model can generalize effectively across different cyber threats, making it adaptable and reliable in detecting diverse and evolving malware threats. The emphasis on diversity and representativeness in both fields helps create models that are better prepared for real-world challenges.

2. Temporal Data and Sequence Modeling

In both atmospheric sciences and weather prediction, time-series data is essential for modeling how conditions evolve over time, whether hourly, daily, or seasonally. This sequential data helps models accurately predict future weather patterns by understanding changes in variables like temperature, pressure, and wind speed. Similarly, a good malware dataset should capture the temporal sequence of system events, network activity, and user behavior to model the progression of an attack. By tracking how malware behaviors unfold over time—from initial infiltration to exploitation—these datasets enable real-time threat detection and help models adapt to evolving attack patterns.

3. Handling Rare and Extreme Events

In both atmospheric sciences and weather prediction, it is crucial to include rare but significant events, such as hurricanes, tornadoes, or flash floods, in datasets. These extreme occurrences can greatly influence predictions and outcomes, so capturing them ensures models are not just trained on everyday weather patterns but can also accurately predict and respond to high-impact situations. Similarly, a **good malware dataset** should encompass rare but critical attack types, such as advanced persistent threats (APTs) or zero-day exploits. Although these attacks are infrequent, their potential impact is substantial. Including such rare events in the dataset ensures that models are robust and equipped to detect unusual but highly consequential threats, thereby enhancing their resilience and real-world applicability.

4. Feature Selection and Importance

In both atmospheric sciences and weather prediction, selecting key variables like temperature, wind speed, and humidity is essential for accurate predictions, as the choice of features directly influences the quality of the model. Identifying the most predictive atmospheric variables, such as air pressure changes for storm forecasts, refines the models and improves their performance. Similarly, a good malware dataset requires careful selection of important features, such as API calls, network traffic patterns, and system events. Focusing on the most relevant features enhances the effectiveness of detection models by improving accuracy and reducing noise, making them more capable of identifying threats.

5. Benchmarking and Standardization

In both atmospheric sciences and weather prediction, benchmark datasets and standardized evaluation metrics are essential for consistently assessing model accuracy and reliability across different conditions. Standardized weather datasets, such as those from the European Centre for Medium-Range Weather Forecasts (ECMWF), enable meaningful comparisons of forecasting approaches. Similarly, in malware research, having benchmark datasets allows for fair comparisons of detection techniques under consistent conditions. This standardization helps improve model development by providing a clear basis for evaluating and refining different detection approaches, ensuring that models are robust and effective across various scenarios.

6. Adaptability and Update Frequency

In both atmospheric sciences and weather prediction, regularly updating datasets is crucial to account for evolving climatic conditions and new data sources, such as satellite observations, ensuring models remain accurate and adaptable to changes. Similarly, for a good malware dataset, frequent updates are essential to incorporate new malware types, evolving attack methods, and emerging security threats. This continuous updating process helps maintain the effectiveness of detection models, enabling them to adapt to the rapidly changing cyber threat landscape and ensuring robust, real-time threat detection.

7. Data Quality and Integrity

In both atmospheric sciences and weather prediction, ensuring high data quality is essential for accurate model predictions. This involves reducing noise, correcting errors, and ensuring data completeness through reliable data collection methods, such as satellite observations and ground stations. Similarly, for a good malware dataset, maintaining data quality means filtering out noise, ensuring accurate labeling, and minimizing false positives. High data integrity is crucial in both fields, as it directly affects the reliability and trustworthiness of the models, enabling them to make accurate and effective predictions or detections.

Energy consumption

This section focuses on leveraging temporal data, behavioral insights, and diverse datasets to enhance anomaly detection and model performance in energy systems.

1. Feature Correlations and Importance:

Energy: There are strong correlations between input features, and individual features have varying impacts on the model. Feature selection methods like Gradient Boosting Classifier are crucial for these tasks.

Malware: This is similar to the behavior of malware detection datasets, where different features (e.g., system calls, network activity) have varying importance, and feature selection is vital for detecting malicious behavior efficiently.

2. Time Interval and Energy Source:

Energy: The dataset needs to include a time interval or range, and different methods of energy generation (e.g., gas turbine, steam turbine) should be represented.

Malware: For malware detection, data also benefits from temporal information, like time of activity (e.g., sequence of system calls over time), and the type of malicious activity (e.g., fileless, trojan, ransomware) impacts the detection model.

3. Environmental and External Factors:

Energy: External factors like temperature, wind speed, and rainfall affect energy consumption, and the location of the building is linked to these weather variables.

Malware: External factors (e.g., network environment, system vulnerabilities) and context (e.g., user behavior, geographical location of attacks) can influence the likelihood of malware execution. Incorporating environmental data could improve detection accuracy.

4. Dataset Variety and IoT Use:

Energy: Datasets vary in region, sampling rate, collection period, and device usage (e.g., IoT devices). They also differ in accessibility (private or public). There are gaps in datasets from certain regions, such as Africa, which affects generalizability.

Malware: Malware datasets also vary in data sources (e.g., network logs, system logs, IoT devices). The diversity in dataset characteristics (e.g., types of malware, environments, and platforms) can limit generalization, similarly to how energy datasets may be region-specific or influenced by different IoT devices.

5. Behavioral Data and Human Factors:

Energy: To improve energy consumption predictions, datasets should also record occupant behavior, offering deeper insights into energy usage patterns.

Malware: The concept of incorporating behavioral data from users (e.g., patterns of normal system usage) aligns with malware detection efforts to detect anomalous behaviors, such as unexpected file system access or unusual network traffic, which indicate potential malware activity.

6. Training and Prediction Performance:

Energy: Data-driven energy consumption models often perform better in training due to limited, unpredictable data in certain regions or homes, which makes it difficult to generalize.

Malware: Similarly, malware detection models tend to perform better in training when datasets are richer and include more varied attack types. However, the limited availability of labeled data (e.g., samples of new malware variants) can restrict generalization and predictive power.

Hardware

This section explores key aspects of hardware-based anomaly detection, focusing on the importance of diverse datasets, feature relevance, temporal context, accurate labeling, platform diversity, and support for both signature-based and anomaly-based detection methods to enhance malware analysis

1. Representative and Diverse Data

- **Correlation:** Hardware papers emphasize capturing a wide variety of system behaviors (normal and malicious) using hardware performance counters (HPCs) and diverse attack vectors.
- **Transferable Insight:** A good malware dataset must encompass diverse malware types, behaviors, and environments. For example:
 - Include different attack phases (exploitation, payload delivery, system takeover).
 - Incorporate benign behaviors from various software and system states to differentiate normalcy from anomalies effectively.

2. Feature Quality and Relevance

- **Correlation:** The hardware papers discuss extracting relevant features such as microarchitectural events (cache misses, branch mispredictions) to improve malware detection.
- **Transferable Insight:**
 - A malware dataset should focus on collecting high-quality, discriminative features that can capture subtle differences between benign and malicious activity.
 - Features like system call patterns, execution flow anomalies, or memory access behaviors should be prioritized.

3. Temporal and Contextual Information

- **Correlation:** Hardware anomaly detection relies on observing behavioral changes over time, leveraging sampling at specific intervals to detect deviations.
- **Transferable Insight:**
 - A malware dataset should include temporal context, capturing data at various time intervals to facilitate dynamic and behavioral analysis.
 - This enables the detection of evolving threats and zero-day attacks by modeling temporal deviations from normalcy.

4. Labeling and Ground Truth

- **Correlation:** hardware papers stress the importance of accurate labeling of data, including stages of malware execution and clean versus infected states.
- **Transferable Insight:**
 - Datasets must include well-labeled samples with clear distinctions between benign and malicious activities.
 - Labels should cover diverse malware categories (e.g., ransomware, spyware, rootkits) and multiple stages of infection.

5. Diverse Platforms and Environments

- **Correlation:** The hardware papers use specific platforms (e.g., Intel processors, Windows OS) to detect anomalies but suggest the need for broader applicability.

- **Transferable Insight:**
 - A malware dataset should include samples from diverse platforms, operating systems, and hardware configurations to improve generalizability across detection systems.
 - This diversity helps mitigate biases and ensures robust performance across varied use cases.

6. Support for Anomaly Detection

- **Correlation:** hardware papers highlight the effectiveness of anomaly-based detection, especially for identifying unknown or zero-day threats.
- **Transferable Insight:**
 - A malware dataset should be designed to support both signature-based and anomaly-based detection systems.
 - Including sufficient benign examples and edge cases can help models learn what constitutes normal behavior, thereby improving anomaly detection.

7. Reproducibility and Standardization

- **Correlation:** The use of standardized tools (e.g., Metasploit, HPCs) in these studies ensures reproducibility and comparability.
- **Transferable Insight:**
 - A malware dataset should be created using standardized tools and methodologies, enabling reproducibility and consistent benchmarking.
 - Public availability and documentation of the dataset enhance its utility in academic and industry settings.

8. Feature Selection and Dimensionality Reduction

- **Correlation:** The papers employ methods like principal component analysis (PCA) and filtering algorithms to select the most relevant hardware events for detection.
- **Transferable Insight:**
 - Malware datasets should integrate techniques for feature selection to ensure that collected features are meaningful and computationally efficient for detection systems.
 - This reduces noise and enhances model performance by focusing on impactful attributes.

9. Support for Machine Learning

- **Correlation:** Both papers leverage machine learning techniques (e.g., one-class SVMs, supervised classification) for malware detection.
- **Transferable Insight:**
 - A malware dataset should be structured to support both supervised and unsupervised machine learning techniques.

- Data should be balanced and sufficient to train robust models, including support for deep learning approaches.

10. Handling Evasion Techniques

- **Correlation:** The hardware papers address challenges like mimicry attacks and stealthy malware that evade detection by mimicking normal behavior.
- **Transferable Insight:**
 - A malware dataset should include samples designed to simulate evasion techniques (e.g., polymorphic or metamorphic malware) to test detection resilience.
 - This ensures models are robust against adversarial tactics and real-world complexities.

6. Issues and Challenges

This section addresses key challenges in malware dataset development, including the lack of standardized benchmarks, difficulties in adapting cross-disciplinary practices, scalability issues, the dynamic nature of emerging threats, and the limited generalizability of dataset characteristics across diverse use cases.

- **Lack of Standardized Benchmarks:** The lack of standardized benchmarks makes it difficult to evaluate and compare different malware datasets consistently. Without common evaluation metrics, determining which characteristics truly contribute to dataset quality becomes subjective.
- **Challenges in Adopting Cross-Disciplinary Best Practices:** While insights from fields like weather prediction and image processing are valuable, adapting their best practices to malware datasets presents unique challenges due to fundamental differences in data structure and characteristics.
- **Scalability Limitations:** Methods that work effectively for analyzing smaller datasets often face significant challenges when applied to larger-scale malware collections. This scalability issue affects both the analysis process and the validation of findings.
- **Accounting for Emerging Threats:** The constant evolution of malware threats makes it challenging to ensure that identified characteristics remain relevant and applicable to emerging threats and attack patterns.
- **Generalizability of Characteristics:** Identifying which characteristics of a dataset can be consistently applied across various dataset types and use cases is challenging, as different datasets may have unique properties, biases, or specific features that limit the generalization of findings.

7. Conclusions and Recommendations

1. Diversity and Representativeness

A malware dataset must encompass a wide range of attributes to reflect real-world threats accurately.

- **Malware Type Coverage:** Include diverse malware families such as ransomware, trojans, worms, rootkits, APTs (Advanced Persistent Threats), and zero-day exploits. These samples should capture behaviors unique to each family.
- **Environmental Representation:** Ensure datasets include samples from various environments, including corporate, cloud, consumer, and IoT settings. Coverage should extend across multiple operating systems (Windows, Linux, macOS) and architectures (Intel, ARM).
- **Rare and High-Impact Events:** Similar to atmospheric sciences where rare events like hurricanes are crucial, include low-frequency but impactful malware types such as advanced obfuscation methods or nation-state-level attacks.
- **Diverse Attack Strategies:** Capture a range of attack types such as phishing, SQL injection, credential stuffing, and Distributed Denial of Service (DDoS). IoT datasets emphasize the importance of traffic balance; malware datasets should achieve similar diversity.

Impact: Diverse datasets ensure that machine learning models are equipped to generalize across unseen threats and reduce biases stemming from overrepresentation of common samples.

2. Temporal Coherence and Realism

Datasets must reflect the chronological and behavioral progression of malware to ensure real-world applicability.

- **Chronological Training and Testing:** Training data should strictly predate testing data. This practice, akin to weather prediction datasets, avoids artificially inflated model performance by simulating real-world scenarios.
- **Lifecycle Modeling:** Capture temporal sequences, such as how malware infiltrates, spreads, and executes payloads. Temporal coherence allows models to detect evolving threats in real time.
- **Avoid Random Splits:** Random sampling of training and testing data should be avoided as it introduces data leakage, as noted in IoT dataset evaluations. Time-based splits provide a more accurate reflection of real-world detection challenges.

Impact: Temporal alignment allows for effective prediction and mitigation of evolving malware behaviors, improving model robustness in live environments.

3. Data Quality and Integrity

The reliability of the dataset depends on its consistency, accuracy, and noise-free nature.

- **Annotation Precision:** Labels for malware types and behaviors should be accurate and consistent. Incorrect labeling increases false positives and false negatives, which are particularly critical in cybersecurity applications.
- **Noise Reduction:** IoT datasets demonstrate the importance of cleaning data by removing irrelevant features. For malware datasets, eliminating redundant or misleading attributes (e.g., benign activity misclassified as malicious) ensures better training results.

Impact: High-quality datasets improve trust in model predictions and facilitate consistent results across different research contexts.

4. Adaptability and Scalability

As malware evolves rapidly, datasets must be dynamic and scalable to remain effective over time.

- **Frequent Updates:** Borrowing from IoT datasets that use long-term monitoring, malware datasets should be updated periodically to include the latest threats, especially emerging tactics and tools.
- **Dynamic Corpus Management:** Use web scraping, crowdsourcing, or honeypots to enrich the dataset with real-time malware samples.
- **Scalable Architectures:** Design datasets that can grow without compromising data quality or model performance, similar to graph-based datasets that scale from small graphs to structures with millions of nodes.

Impact: Dynamic updates ensure datasets remain relevant, helping models to adapt to the evolving nature of malware threats.

5. Feature Selection and Augmentation

Choosing the right features significantly impacts the dataset's utility in training and testing machine learning models.

- **Relevant Features:** Prioritize features like API calls, system events, network traffic patterns, and file attributes. This mirrors practices in IoT datasets, which focus on session information and protocol-specific flags.
- **Augmented Features:** Use feature engineering to derive high-value attributes, such as correlation between API calls or time delays in network activity.
- **Custom Features for Specific Scenarios:** Design features tailored to rare malware behaviors, like polymorphic or fileless attacks, to ensure robust detection.

Impact: Effective feature selection improves model accuracy and reduces computational overhead by eliminating irrelevant information.

6. Realism and Practical Application

Datasets must simulate real-world scenarios to provide meaningful insights and ensure practical applicability.

- **Simulated but Realistic Scenarios:** Mimic real-world conditions by introducing noise, mixed traffic, and edge cases.
- **Behavioral Validation:** Incorporate dynamic analysis data, such as malware execution traces or system call monitoring, to capture runtime behaviors. This mirrors IoT practices where both flow-level and protocol-level data are analyzed.
- **Cross-Dataset Generalization:** As seen in image processing, ensure models trained on the dataset perform well on other datasets, demonstrating its representativeness and robustness.

Impact: Realistic datasets bridge the gap between research and real-world deployments, making findings more actionable.

7. Standardized Benchmarks and Evaluation

A unified evaluation framework ensures consistency in assessing dataset quality and model performance.

- **Evaluation Protocols:** Borrowing from weather prediction, define clear benchmarks and evaluation metrics like detection accuracy, precision, recall, and false positive rates.
- **Reproducibility:** Ensure public availability of datasets and fixed data splits to allow for consistent comparisons across studies.
- **Custom Metrics:** Introduce context-specific metrics, such as resilience to concept drift (i.e., changes in malware behavior over time).

Impact: Standardized benchmarks enable meaningful comparisons and promote rigorous evaluation of datasets and detection techniques.

8. Insights from Cross-Industry Practices

Integrating best practices from other domains enhances the quality and utility of malware datasets.

- **IoT and Network Security:**
 - Emphasize traffic balance and protocol-specific features to simulate diverse attack environments.
 - Adopt ensemble-based approaches to evaluate datasets for anomaly detection.
- **Weather Prediction:**

- Include rare but impactful events and leverage time-series data for sequential modeling of malware behaviors.
- **Image Processing:**
 - Reduce dataset biases through diverse sampling and include realistic negative samples for comprehensive training.
- **Graph-Based Models:**
 - Ensure scalability and structural diversity in dataset design, enabling more generalized approaches to malware detection.

Impact: Cross-industry insights help tackle unique challenges in malware dataset construction, promoting innovation and robustness.

Conclusion

A high-quality malware dataset is defined by its ability to accurately represent the diversity, complexity, and evolving nature of malware threats. By incorporating lessons from various industries, we recommend:

1. Emphasizing **diversity** to capture a wide spectrum of malware behaviors.
2. Incorporating **temporal coherence** to reflect real-world conditions.
3. Ensuring **high data quality** to maintain trust and consistency in research.
4. Designing for **adaptability and scalability** to address emerging threats.
5. Focusing on **realism and standardization** to bridge the gap between research and operational applications.

These recommendations aim to create datasets that are not only scientifically rigorous but also operationally effective, enabling the cybersecurity community to stay ahead of evolving threats. By following this framework, researchers can contribute to more accurate detection systems and stronger defenses against malware.

8. Future Work

Future work could investigate leveraging AI/ML techniques, including deep learning, reinforcement learning, and unsupervised learning, to improve the process of dataset construction. Additionally, research could explore the use of graph-based models to capture interactions between malware and system components. To make the dataset more realistic and challenging for detection systems, it would be beneficial to include malware samples that employ advanced evasion techniques, such as polymorphism and anti-sandbox behaviors. Moreover, automated systems could be developed to continuously collect and update malware samples in real time using honeypots, malware sandboxes, and threat intelligence feeds. We can further explore various industries to uncover new methods for structuring and organizing malware datasets. By incorporating data architectures and analytical techniques from fields such as medical imaging,

autonomous systems, or financial fraud detection, we can create more robust, dynamic datasets that better capture the complexities of malware behavior.

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10. Team Biography / Tasking

Mohammad Rafieian is a Ph.D. candidate in Software Engineering at The University of Texas at Dallas, with a focus on software security and machine learning. His research aims to enhance the efficiency of automated program analyzers in detecting vulnerabilities by applying advanced machine learning methods. As a Research Assistant, Mohammad has contributed to the development and testing of innovative program analysis tools, playing a pivotal role in advancing cybersecurity. In addition to his research, he has worked as a Backend Developer, where he designed scalable architectures and optimized system performance. Proficient in Python, Java, and Docker, Mohammad's combination of cutting-edge security research and hands-on development experience makes him a valuable contributor to any cybersecurity project.

Tharun Mulka is a second-year Ph.D. student at The University of Texas at Dallas, specializing in cybersecurity, with a strong foundation in both academic research and practical applications. He holds a bachelor's degree in computer science, where his research on email security led to the discovery of vulnerabilities in the DKIM authentication protocol, sharpening his problem-solving and analytical skills. His experience publishing research has strengthened his academic writing, while his advanced coursework in Network Security, Algorithms, and Information Security has deepened his technical knowledge. Tharun has demonstrated his ability to build complex systems from scratch, such as implementing end-to-end encryption for an e-commerce platform without relying on external libraries, showcasing his expertise in cryptography. His current Ph.D. research focuses on securing data and privacy against USB-based attacks, which has given him hands-on experience with the USB protocol and further enhanced his coding skills. Given his diverse experience and ability to quickly master new concepts, Tharun is well-equipped to excel in his project and contribute effectively to the team's success.

Sahithi Reddy Mopuru is currently pursuing her Master of Science in Computer Science at The University of Texas at Dallas, building on a strong foundation established during her undergraduate studies in computer science and engineering at SRM University, Chennai. With an impressive academic background, Sahithi has participated in several significant research projects that have honed her technical and analytical skills. During her bachelor's, she developed a keen interest in data science and machine learning, which culminated in a project

focused on detecting counterfeit medical remedies using an LSTM model in the TensorFlow framework. This work, presented at the IEEE 12th International Conference on Data Science and Engineering, gave her valuable insights into healthcare misinformation and highlighted her commitment to leveraging technology for societal impact. In her graduate studies, Sahithi has expanded her expertise in data science, excelling in data analysis, machine learning, and statistical modeling with tools like Python, TensorFlow, and R. Her technical skills were further demonstrated through her development of a smart fish farming recommendation system, which used reinforcement learning algorithms to optimize feeding schedules and enhance fish health. In addition to her work in data science, Sahithi has cultivated a strong interest in cybersecurity, focusing on understanding and mitigating digital threats. This combination of data science and cybersecurity skills equips her to solve complex, real-world problems with innovative solutions. Currently, as a Grading Assistant at UT Dallas, Sahithi continues to refine her technical abilities by providing data analysis support and personalized feedback to students, contributing to both her research pursuits and professional growth.

Sri Kruthi Madugula is a Master of Science student in Computer Science with a specialization in Data Science at The University of Texas at Dallas. Her academic background, combined with hands-on experience, makes her a valuable contributor to cybersecurity projects. Sri Kruthi has developed a strong foundation in data structures, algorithms, machine learning, and big data management—skills essential for analyzing and mitigating cyber threats. Her proficiency in designing and implementing complex algorithms is especially relevant for building robust malware detection systems that require advanced data processing and analysis to identify potential security breaches. During her tenure as a Data Engineering Intern at Tenet HealthCare, Sri Kruthi gained significant experience managing and securing large-scale data environments. She successfully migrated SQL Server stored procedures to Google BigQuery, demonstrating her expertise in handling sensitive data, ensuring security, and optimizing systems for real-time analytics—skills crucial for maintaining data integrity in cybersecurity. Her work on machine learning models for liver disease prediction and grape disease detection further highlights her ability to apply advanced analytics to solve complex problems. In the cybersecurity context, these abilities are transferable to developing predictive models for threat detection and response. With a combination of strong academic knowledge and practical engineering experience, Sri Kruthi brings a unique blend of data science expertise to any cybersecurity-focused team. Her capabilities in data-driven approaches make her a key player in projects aimed at enhancing digital security.

Pranathi Munnangi is a second-year Master's student in Computer Science, specializing in Data Science. She earned her Bachelor's degree in Computer Science, with a focus on cloud computing, cybersecurity, network security, and AI. This strong foundation has provided her with a deep understanding of securing systems, protecting sensitive data, and deploying secure cloud infrastructures. In her current studies, Pranathi has developed expertise in advanced data science techniques, including machine learning, data mining, and big data technologies. She is proficient in programming languages such as Python, R, C++, and Java, and has experience working with SQL and Spark for data processing and analysis. Her interest in cloud computing

has led her to explore the integration of AI and machine learning in cloud environments, with a focus on optimizing performance and enhancing security. In her project, "Generating Images Using Complex Sentences," Pranathi investigated the capabilities of Stable Diffusion models, a cutting-edge technology in generative AI. This diverse background in AI, cloud computing, and cybersecurity, combined with her practical programming and networking experience, equips Pranathi to address challenges at the intersection of these fields. Pranathi is dedicated to integrating robust safety protocols into AI models and cloud systems, ensuring that these technologies operate reliably and ethically. Her ultimate goal is to develop AI and cloud computing tools that enhance human capabilities while minimizing potential risks, reflecting her commitment to creating a more secure and trustworthy technological future.

Appendix A: Acronyms and Terms / Definitions

PE : Portable Executable File

APT: Advanced Persistent Threat

OGB: Open Graph Benchmark

TCP/IP: Transmission Control Protocol/Internet Protocol

RNN: Recurrent Neural Network

DKIM: DomainKeys Identified Mail

SMOTE: Synthetic Minority Over-sampling Technique

QOS: Quality of Service

PCA: Principal Component Analysis

HPC: Hardware Performance Counters

MQTT protocol: Message Queuing Telemetry Transport is a lightweight, publish-subscribe protocol designed for efficient communication in IoT and low-bandwidth networks.

Malware: Malicious software designed to disrupt, damage, or gain unauthorized access to computer systems.

Dataset Bias: The presence of systematic errors or skewness in data collection, leading to a dataset that does not fully represent the diversity or complexity of real-world malware.

Dataset Quality: The criteria by which a dataset is evaluated, including factors such as accuracy, completeness, consistency, and relevance.

Real-world Malware: Malware samples that are observed in actual cyber-attacks or infections, as opposed to artificial or laboratory-created samples.

Data Diversity: The range and variety of malware samples included in a dataset to ensure a comprehensive representation of different malware types and behaviors.

Malware Detection: The process of identifying malware based on its signatures, behaviors, or characteristics within a system or dataset.

False Positives/Negatives: Incorrect results in malware detection, where benign data is identified as malware (false positive) or malware goes undetected (false negative).

A **representative malware dataset** is a diverse and comprehensive collection of malware samples that accurately reflects the variety and complexity of real-world threats, ensuring effective analysis and detection across different malware types.

Reinforcement Learning: A type of machine learning where an agent learns how to behave in an environment by performing actions and receiving feedback in the form of rewards or penalties.

Metamorphic Malware: A type of malware that rewrites its own code each time it infects a system, making it different from polymorphic malware, which only changes part of its code.